PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 2 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

81 classified items • 317 marks • 18 chapter(s)

Source-faithful practice with synchronized solutions

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ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 2 • Expertise • Unit 2 • Expertise Q16+ • 81 items • 317 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Direct & Inverse Proportion

Unit 2 • Expertise • 19 marks • 5 item(s)

June 2025 | 4MA1-1H Q18 | 4 marks

Recover the input under inverse variation

June 2023 | 4MA1-2H Q16 | 5 marks

Root direct variation y=ksqrtx

June 2022 | 4MA1-2H Q17 | 4 marks

Power direct variation y=kx^n

June 2021 | 4MA1-2H Q16 | 3 marks

Inverse-square or inverse-power variation

June 2021 | 4MA1-2H Q16 | 3 marks

Inverse-square or inverse-power variation

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Algebra Toolkit

Unit 2 • Expertise • 7 marks • 2 item(s)

June 2024 | 4MA1-2H Q20 | 3 marks

Signed, decimal or fractional formula substitution

June 2023 | 4MA1-2H Q17 | 4 marks

Subject appears in two linear terms

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Completing the Square

Unit 2 • Expertise • 3 marks • 2 item(s)

November 2021 | 4MA1-1H Q19 | 2 marks

Negative leading coefficient or maximum form

November 2021 | 4MA1-1H Q19 | 1 marks

Negative leading coefficient or maximum form

Original question context and synchronized companion pages follow.

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Rearranging Formulas

Unit 2 • Expertise • 8 marks • 2 item(s)

June 2025 | 4MA1-2H Q16 | 4 marks

Single squared/powered subject

January 2020 | 4MA1-2H Q16 | 4 marks

Subject in the denominator

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Algebraic Proof

Unit 2 • Expertise • 16 marks • 5 item(s)

January 2022 | 4MA1-2H Q18 | 3 marks

Fixed remainder modulo a divisor

January 2020 | 4MA1-2H Q17 | 3 marks

Consecutive integer parity identity

June 2021 | 4MA1-1H Q17 | 3 marks

Consecutive even/odd sequence proof

November 2021 | 4MA1-1H Q19 | 3 marks

Derive a general geometric algebra formula

January 2023 | 2H Q17 | 4 marks

Product or sum of arbitrary odd/even numbers

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Solving Inequalities

Unit 2 • Expertise • 16 marks • 4 item(s)

June 2025 | 4MA1-2H Q19 | 3 marks

Factorable quadratic sign intervals

January 2022 | 4MA1-2H Q22 | 5 marks

Contextual quadratic feasible range

May-Nov 2020 | 2H Q17 | 5 marks

Inequality symbol from a number line

January 2023 | 2H Q22 | 3 marks

One-sided linear inequality

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Simultaneous Equations

Unit 2 • Expertise • 25 marks • 5 item(s)

June 2024 | 4MA1-2H Q22 | 5 marks

Line with mixed quadratic relation

November 2024 | 4MA1-2H Q22 | 5 marks

Line with circle or sum-of-squares curve

January 2020 | 4MA1-2H Q22 | 5 marks

Multiple solution-pair validation

May-Nov 2020 | 2H Q16 | 5 marks

Draw the missing line for an intersection

January 2023 | 2H Q20 | 5 marks

Draw the missing line for an intersection

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Sequences

Unit 2 • Expertise • 25 marks • 5 item(s)

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Functions

Unit 2 • Expertise • 19 marks • 7 item(s)

June 2025 | 4MA1-1H Q16 | 1 marks

Evaluate a function at a numerical input

June 2025 | 4MA1-1H Q16 | 3 marks

Solve a composite function inequality

June 2023 | 4MA1-1H Q17 | 4 marks

Exclude a denominator zero from the domain

January 2022 | 4MA1-2H Q25 | 1 marks

Domain and range swap under inversion

June 2021 | 4MA1-1H Q24 | 2 marks

Evaluate a numerical composite function

June 2021 | 4MA1-1H Q24 | 4 marks

Inverse of a restricted quadratic function

January 2023 | 1H Q18 | 4 marks

Inverse of a composite-defined function

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Graphs of Functions

Unit 2 • Expertise • 10 marks • 3 item(s)

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Transformations of Graphs

Unit 2 • Expertise • 11 marks • 8 item(s)

June 2024 | 4MA1-2H Q25 | 2 marks

Horizontal translation

November 2024 | 4MA1-2H Q19 | 1 marks

Vertical scale or x-axis reflection

November 2024 | 4MA1-2H Q19 | 1 marks

Vertical translation

June 2022 | 4MA1-2H Q23 | 1 marks

Horizontal scale or y-axis reflection

November 2021 | 4MA1-2H Q21 | 1 marks

Transform an explicit polynomial and simplify

November 2021 | 4MA1-2H Q21 | 1 marks

Translation vector applied to y=f(x)

November 2021 | 4MA1-2H Q21 | 1 marks

Describe from two curve equations

November 2021 | 4MA1-2H Q21 | 3 marks

Combined feature mapping

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Differentiation

Unit 2 • Expertise • 34 marks • 7 item(s)

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Angles in Polygons & Parallel Lines

Unit 2 • Expertise • 6 marks • 1 item(s)

June 2022 | 4MA1-1H Q23 | 6 marks

Solve an irregular polygon angle or side-count problem

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Bearings, Scale Drawing & Constructions

Unit 2 • Expertise • 16 marks • 3 item(s)

November 2023 | 4MA1-1H Q18 | 5 marks

Route geometry followed by distance/time synthesis

November 2024 | 4MA1-2H Q24 | 6 marks

Sine or cosine rule supports a final bearing

January 2021 | 4MA1-1H Q16 | 5 marks

Sine or cosine rule supports a final bearing

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Circle Theorems

Unit 2 • Expertise • 14 marks • 3 item(s)

November 2023 | 4MA1-1H Q22 | 6 marks

Use radius perpendicular to tangent

November 2024 | 4MA1-2H Q16 | 5 marks

Complete a multi-theorem circle-angle chain

January 2022 | 4MA1-2H Q19 | 3 marks

Use the internal intersecting-chords product

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Volume & Surface Area

Unit 2 • Expertise • 28 marks • 6 item(s)

June 2024 | 4MA1-1H Q23 | 5 marks

Find volume or surface after removing a solid

November 2023 | 4MA1-1H Q21 | 5 marks

Find volume of a prism from its cross-section

June 2022 | 4MA1-1H Q22 | 5 marks

Find volume of a joined composite solid

January 2020 | 4MA1-2H Q26 | 6 marks

Find cone surface area or recover its sector

January 2021 | 4MA1-2H Q17 | 4 marks

Find surface area of a cuboid or simple prism

January 2023 | 2H Q21 | 3 marks

Find sphere or hemisphere surface area

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Area & Volume of Similar Shapes

Unit 2 • Expertise • 22 marks • 5 item(s)

June 2025 | 4MA1-2H Q26 | 4 marks

Solve a volume difference or sum for similar solids

November 2023 | 4MA1-1H Q23 | 5 marks

Convert between area and volume ratios

November 2023 | 4MA1-2H Q23 | 5 marks

Use similar solids to solve a frustum problem

January 2022 | 4MA1-1H Q17 | 4 marks

Solve a volume difference or sum for similar solids

January 2023 | 2H Q19 | 4 marks

Solve an area difference or sum for similar figures

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Vectors

Unit 2 • Expertise • 38 marks • 8 item(s)

June 2025 | 4MA1-1H Q23 | 5 marks

Find an intersection vector from two paths

June 2024 | 4MA1-2H Q24 | 6 marks

Find an intersection vector from two paths

November 2023 | 4MA1-2H Q22 | 5 marks

Recover a division ratio using vector parameters

June 2023 | 4MA1-1H Q19 | 5 marks

Recover a division ratio using vector parameters

November 2024 | 4MA1-1H Q24 | 6 marks

Find an intersection vector from two paths

November 2024 | 4MA1-2H Q17 | 3 marks

Find magnitude of a vector

May-Nov 2020 | 2H Q19 | 3 marks

Use vectors in a parallelogram, trapezium or polygon path

January 2023 | 1H Q24 | 5 marks

Find a section or midpoint vector

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

$y = 4$ when $x = 3$

Work out the value of x when $y = 864$

$x =$ ....

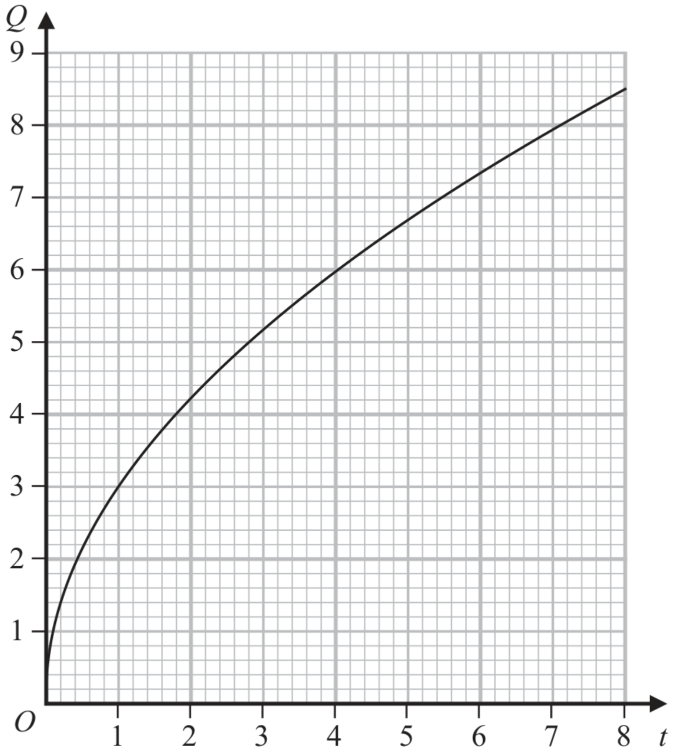
(Total for Question 18 is 4 marks)

WORKING SPACE

4MA1-2H Q16 **5 marks**

16 Q is directly proportional to $\sqrt{t}$

The graph shows the relationship between Q and t for $0 < t < 8$



(a) Find a formula for Q in terms of t

....
(3)

Q is increased by 20%

(b) Find the percentage increase in t

.... %
(2)

Continue your method **5 marks**

WORKING SPACE

4MA1-2H Q17 Q17 **4 marks**

17 M varies directly as the cube of h
$M = 4$ when $h = 0.5$

Find the value of h when $M = 500$

....

WORKING SPACE

16 A is inversely proportional to the square of r

$A = 5$ when $r = 0.3$

(a) Find a formula for A in terms of r

....

(3)

(b) Find the value of A when $r = 7.5A$

$A =$ ....

(3)

YOUR WORKING

16 A is inversely proportional to the square of r

$A = 5$ when $r = 0.3$

(a) Find a formula for A in terms of r

....

(3)

(b) Find the value of A when $r = 7.5A$

$A =$ ....

(3)

YOUR WORKING

4MA1-2H Q20 **3 marks**

20 Given that $k = x - y$ and $x = \dfrac{1}{4y}$

express $\dfrac{5k}{x+2}$ in the form $\dfrac{a - by^2}{c + dy}$ where a, b, c and d are integers.

....

WORKING SPACE

4MA1-2H Q17

4 marks

(b) Make e the subject of $w = \sqrt{\dfrac{e+g}{ef-d}}$

....

(4)

WORKING SPACE

19 *ABCED* is a five-sided shape.

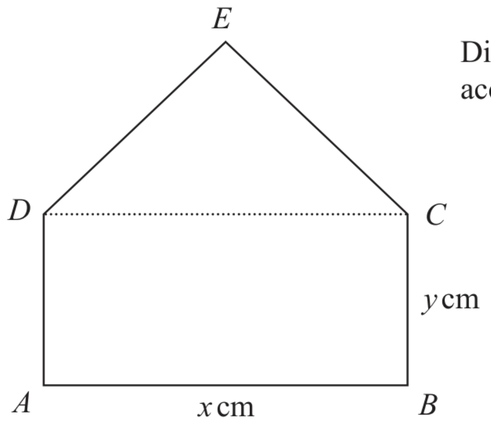


Diagram **NOT** accurately drawn

ABCD is a rectangle.
CED is an equilateral triangle.

$AB = x$ cm $\quad BC = y$ cm

The perimeter of *ABCED* is 100 cm.
The area of *ABCED* is R cm^2

(a) Show that $R = \frac{x}{4}\left(200 - \left[6 - \sqrt{3}\right]x\right)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$x =$ ....

(2)

(ii) Explain why the maximum value of R is given by this value of x.

....

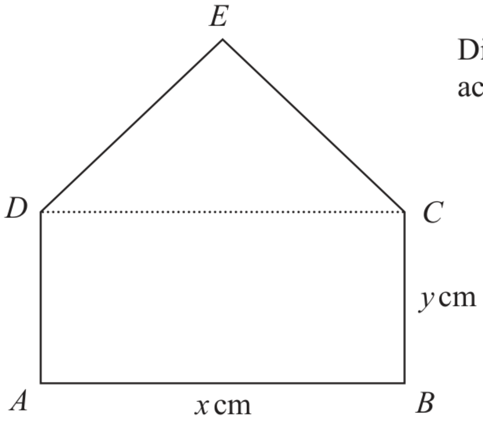
....

....

(1)

YOUR WORKING

19 *ABCED* is a five-sided shape.



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(3)

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Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$x =$ ....

(2)

(ii) Explain why the maximum value of R is given by this value of x.

....

....

....

(1)

YOUR WORKING

$7 - 8t^2$

....

(Total for Question 16 is 4 marks)

WORKING SPACE

January 2020 | 4MA1-2H | Question 16 | 4 marks

Chapter OL-T22: Rearranging Formulas

16 Make x the subject of $y = \sqrt{\dfrac{x+1}{x-4}}$

....

YOUR WORKING

4MA1-2H Q18 **3 marks**

18 Prove that when the sum of the squares of any two consecutive odd numbers is divided by 8, the remainder is always 2
Show clear algebraic working.

WORKING SPACE

January 2020 | 4MA1-2H | Question 17 | 3 marks

Chapter OL-T23: Algebraic Proof

17 Prove that the difference between two consecutive square numbers is always an odd number.
Show clear algebraic working.

YOUR WORKING

17 Using algebra, prove that, given any 3 consecutive even numbers, the difference between the square of the largest number and the square of the smallest number is always 8 times the middle number.

YOUR WORKING

19 *ABCED* is a five-sided shape.

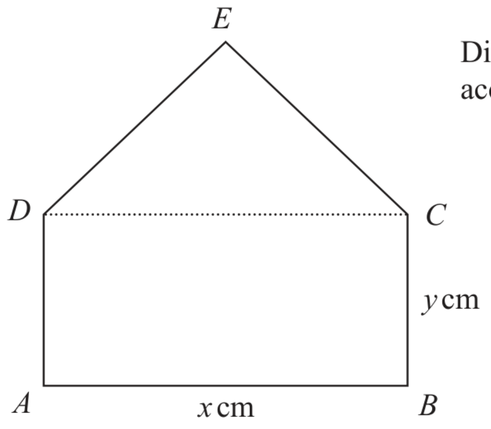

Diagram **NOT** accurately drawn

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(a) Show that $R = \frac{x}{4}\left(200 - \left[6 - \sqrt{3}\right]x\right)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$x =$ ....

(2)

(ii) Explain why the maximum value of R is given by this value of x.

....

....

....

(1)

YOUR WORKING

17 Given that $8\sqrt{m} + \sqrt{49m} - \sqrt{9m} = k\sqrt{m}$

where k is an integer and m is a prime number,

(a) work out the value of k

$k =$ ....

(1)

(b) Show that $\dfrac{5 - \sqrt{18}}{1 - \sqrt{2}}$ can be written in the form $a + b\sqrt{2}$

where a and b are integers.

Show each stage of your working clearly.

(3)

(Total for Question 17 is 4 marks)

YOUR WORKING

Show clear algebraic working.

....

(Total for Question 19 is 3 marks)

WORKING SPACE

4MA1-2H Q22 **5 marks**

22 Here is a rectangle.

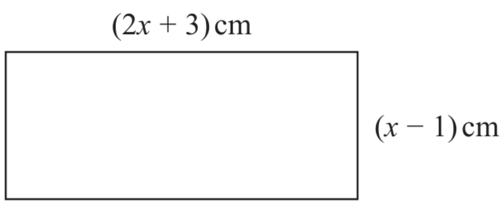


Diagram **NOT** accurately drawn

Given that the area of the rectangle is less than $75\,\text{cm}^2$

find the range of possible values of x

....

WORKING SPACE

17 The diagram shows a rectangle.

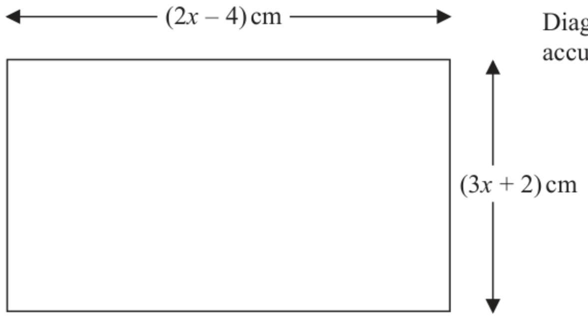


Diagram **NOT** accurately drawn

The area of the rectangle is A cm^2

Given that $A < 3x + 27$
find the range of possible values for x.

....

(Total for Question 17 is 5 marks)

YOUR WORKING

January 2023 | 2H | Question 22 | 3 marks

Chapter OL-T26: Solving Inequalities

22 Solve the inequality $6x^2 + 37x \leqslant 35$
Show clear algebraic working.

....

(Total for Question 22 is 3 marks)

YOUR WORKING

4MA1-2H Q22 **5 marks**

22 The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.
Show clear algebraic working.

(.......................... , ...) (.......................... , ...)

WORKING SPACE

4MA1-2H Q22 M01 **5 marks**

22 Solve the simultaneous equations

$$x^2 + y^2 + y = 3$$

$$x + 2 = y$$

Show clear algebraic working.

....

WORKING SPACE

January 2020 | 4MA1-2H | Question 22 | 5 marks

Chapter OL-T27: Simultaneous Equations

22 The line with equation $y = x + 2$ intersects the curve with equation $x^2 + y^2 - 2y = 24$ at the points A and B.

Find the coordinates of A and B.
Show clear algebraic working.

(.............................. , ...)

(.............................. , ...)

YOUR WORKING

May-Nov 2020 | 2H | Question 16 | 5 marks

Chapter OL-T27: Simultaneous Equations

16 Solve the simultaneous equations

$$3xy - y^2 = 8$$
$$x - 2y = 1$$

Show clear algebraic working.

....

(Total for Question 16 is 5 marks)

YOUR WORKING

20 Solve the simultaneous equations

$$y = 7 - 2x$$
$$x^2 + y^2 = 34$$

Show clear algebraic working.

....

(Total for Question 20 is 5 marks)

YOUR WORKING

4MA1-1H Q24 **6 marks**

24 A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84°
The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.
Show clear algebraic working.

....°

WORKING SPACE

4MA1-1H Q19 demand **6 marks**

19 Here are the first 4 terms in an arithmetic sequence.

3 7 11 15

The last term of the sequence is x
The sum of the terms of the sequence is 7260

Find the value of x
Show clear algebraic working.

$x =$ ....

WORKING SPACE

4MA1-1H Q19 M01 **5 marks**

19 An arithmetic series has first term a and common difference d

The sum of the first 30 terms of the arithmetic series is 4395

The sum of the 10th term and the 20th term is 284

Work out the sum of the first 45 terms of the arithmetic series.
Show clear algebraic working.

....

WORKING SPACE

24 An arithmetic series has first term a and common difference d.

The sum of the first $2n$ terms of the series is four times the sum of the first n terms of the series.

Find an expression for a in terms of d.
Show your working clearly.

$a =$ ....

YOUR WORKING

May-Nov 2020 | 1H | Question 24 | 4 marks

Chapter OL-T29: Sequences

24 Here are the first five terms of an arithmetic sequence.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

....

(Total for Question 24 is 4 marks)

YOUR WORKING

$g(x) = 4x^2$

(a) Find f(9)

....

(1)

WORKING SPACE

$g(x) = 4x^2$

(b) Solve $fg(x) < 0$
Show clear algebraic working.

....
(3)

WORKING SPACE

4MA1-1H Q17 **4 marks**

17 The functions g and h are such that

$$g(x) = \frac{11}{2x - 5}$$

$$h(x) = x^2 + 4 \qquad x \geqslant 0$$

(a) What value of x must be excluded from any domain of g?

....

(1)

(b) Solve $gh(x) = 1$

....

(3)

WORKING SPACE

4MA1-2H Q25 Q25(b) **1 marks**

(b) State the domain of g^{-1}

....
(1)

WORKING SPACE

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \qquad \text{where } x \geqslant 1$$

$$g(x) = 7x - 6$$

(a) Find fg(2)

....

(2)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) =$ ....

(4)

YOUR WORKING

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \qquad \text{where } x \geqslant 1$$

$$g(x) = 7x - 6$$

(a) Find fg(2)

....

(2)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) =$ ....

(4)

YOUR WORKING

18 The function f is such that $f(x) = \dfrac{k}{x}$ where $x \neq 0$ and k is an integer.

(a) Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) =$ ....

(1)

The function g is such that $g(x) = 2 - 3x^4$ where $x \neq 0$

The function h is such that $h(x) = \dfrac{3x}{2 - x}$ where $x \neq 2$

(b) (i) Find $g(-2)$

....

(1)

(ii) Express the composite function hg in the form $hg(x) = \ldots$
Give your answer in its simplest form.

$hg(x) =$ ....

(2)

(Total for Question 18 is 4 marks)

YOUR WORKING

4MA1-1H Q23 M01 **5 marks**

23 The curve **C** has equation $y = x^2 - 8x - 9$
The straight line **L** has equation $y = k$ where k is an integer.

C and **L** intersect at the points A and B

The coordinates of point A are (p, k)
The coordinates of point B are (q, k)

Given that $p - q = 14$

find the value of k
Show clear algebraic working.

4MA1-1H Q23 M01

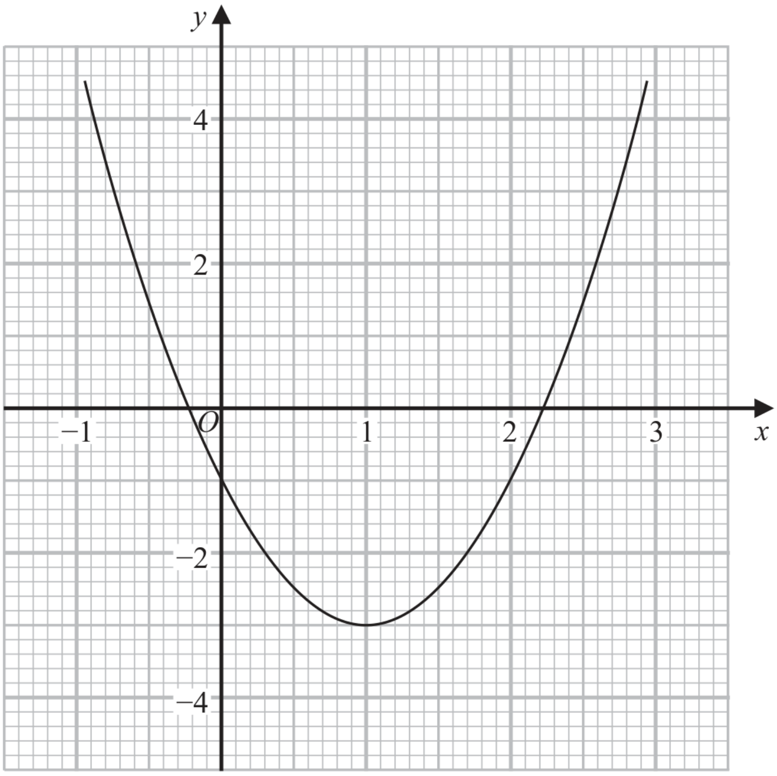
5 marks

$k =$ ....

WORKING SPACE

4MA1-2H Q21 Q21(a) **2 marks**

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



(a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$
Give your solutions correct to one decimal place.

....
(2)

Continue your method **2 marks**

WORKING SPACE

4MA1-2H Q21 Q21(b) **3 marks**

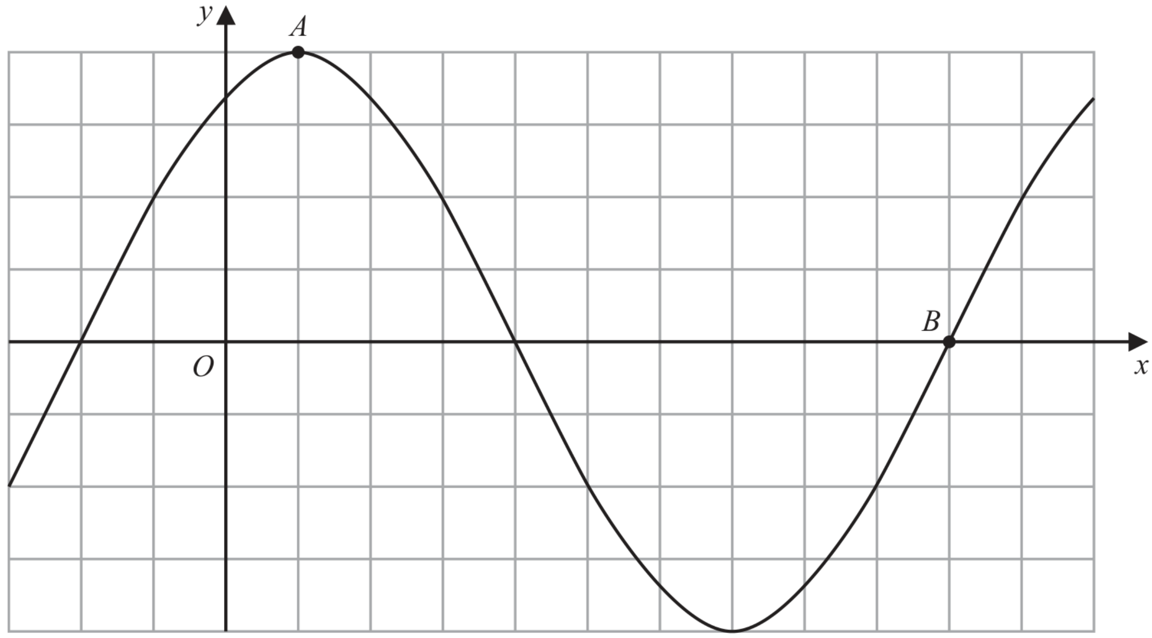
(b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

....

(3)

4MA1-2H Q25 **2 marks**

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)°$



(i) Find the coordinates of the point A

(.............................. , ...)
(1)

(ii) Find the coordinates of the point B

(.............................. , ...)
(1)

Continue your method **2 marks**

WORKING SPACE

4MA1-2H Q19 M02 **1 marks**

19 A curve has equation $y = \mathrm{f}(x)$

There is only one minimum point on the curve.
The coordinates of this minimum point are (5, 4)

(ii) $y = 3\mathrm{f}(x)$

(.............................. , ...)
(1)

WORKING SPACE

4MA1-2H Q19 M03 **1 marks**

19 A curve has equation $y = \mathrm{f}(x)$

There is only one minimum point on the curve.
The coordinates of this minimum point are (5, 4)

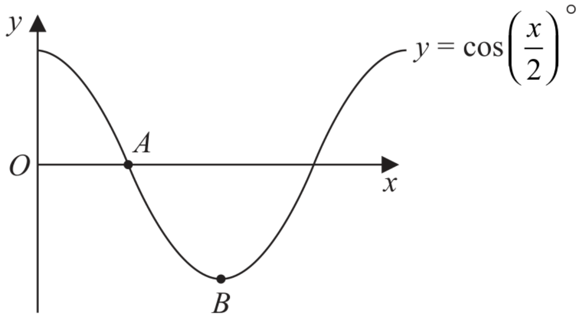
(iii) $y = \mathrm{f}(x) - 7$

(.............................. , ...)
(1)

WORKING SPACE

4MA1-2H Q23 Q23(ii) **1 marks**

23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$



(ii) Find the coordinates of the point B

(.............................. , ...)

(1)

Continue your method **1 marks**

WORKING SPACE

21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point A is the maximum point on **C**.

(a) Write down the coordinates of A.

(.......................... , ...)

(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

....

(1)

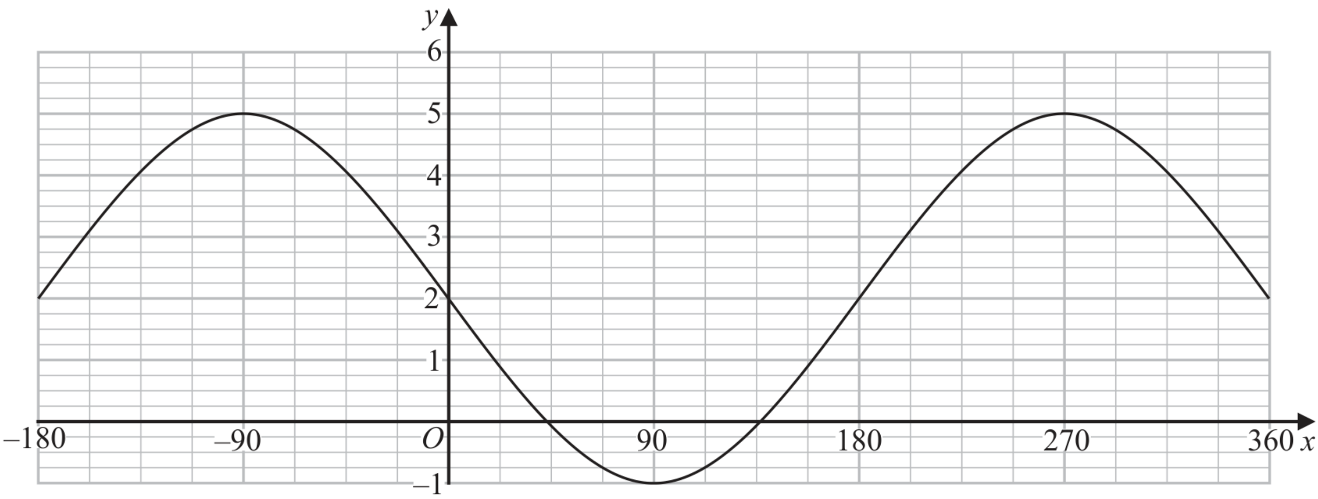
The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

....

(1)

The graph of $y = a\cos(x - b)° + c$ for $-180 \leqslant x \leqslant 360$ is drawn on the grid below.



(d) Find the value of a, the value of b and the value of c.

a = ...

b = ...

c = ...

(3)

YOUR WORKING

21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point A is the maximum point on **C**.

(a) Write down the coordinates of A.

(.............................. , ...)

(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

....

(1)

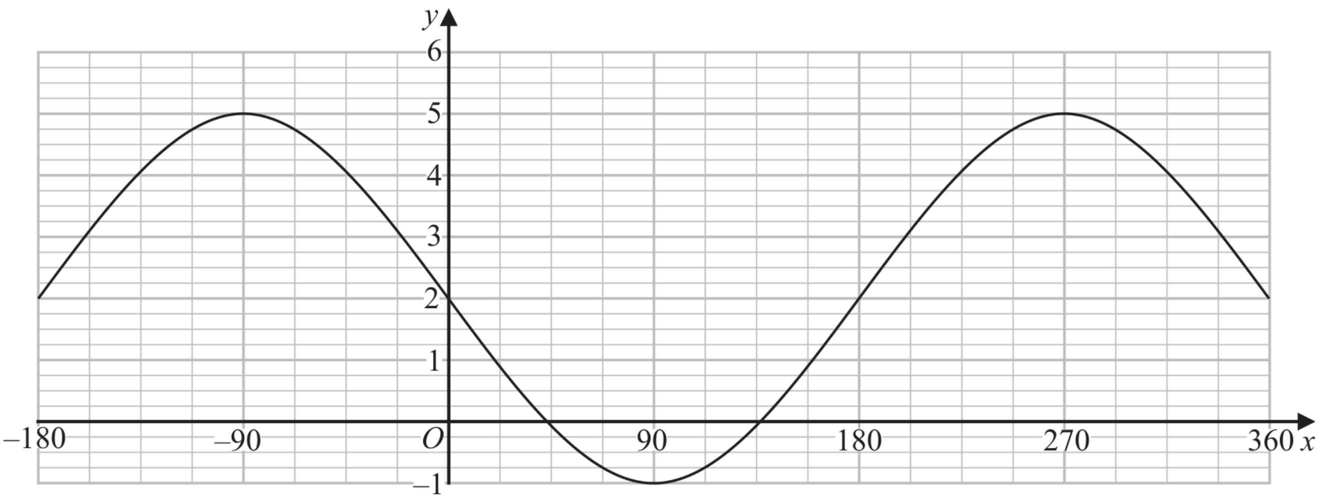
The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

....

(1)

The graph of $y = a\cos(x - b)° + c$ for $-180 \leqslant x \leqslant 360$ is drawn on the grid below.



(d) Find the value of a, the value of b and the value of c.

a = ...

b = ...

c = ...

(3)

YOUR WORKING

21 The curve **C** has equation $y = \text{f}(x)$ where $\text{f}(x) = 9 - 3(x + 2)^2$
The point A is the maximum point on **C**.

(a) Write down the coordinates of A.

(.......................... , ...)

(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

....

(1)

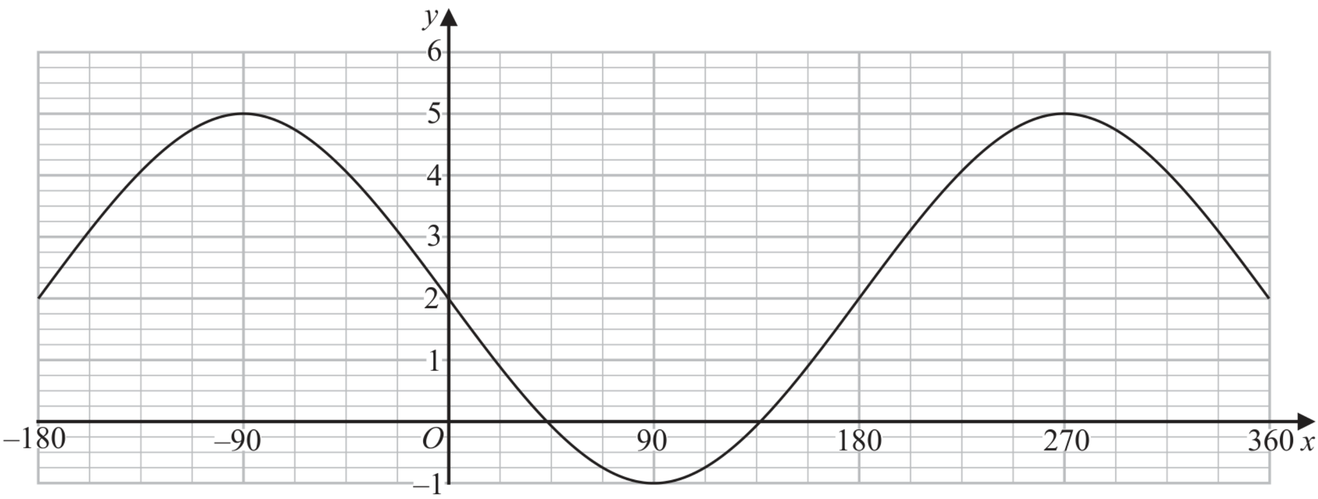
The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

....

(1)

The graph of $y = a\cos(x - b)° + c$ for $-180 \leqslant x \leqslant 360$ is drawn on the grid below.



(d) Find the value of a, the value of b and the value of c.

a = ...

b = ...

c = ...

(3)

YOUR WORKING

21 The curve **C** has equation $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = 9 - 3(x + 2)^2$
The point A is the maximum point on **C**.

(a) Write down the coordinates of A.

(.......................... , ...)

(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

....

(1)

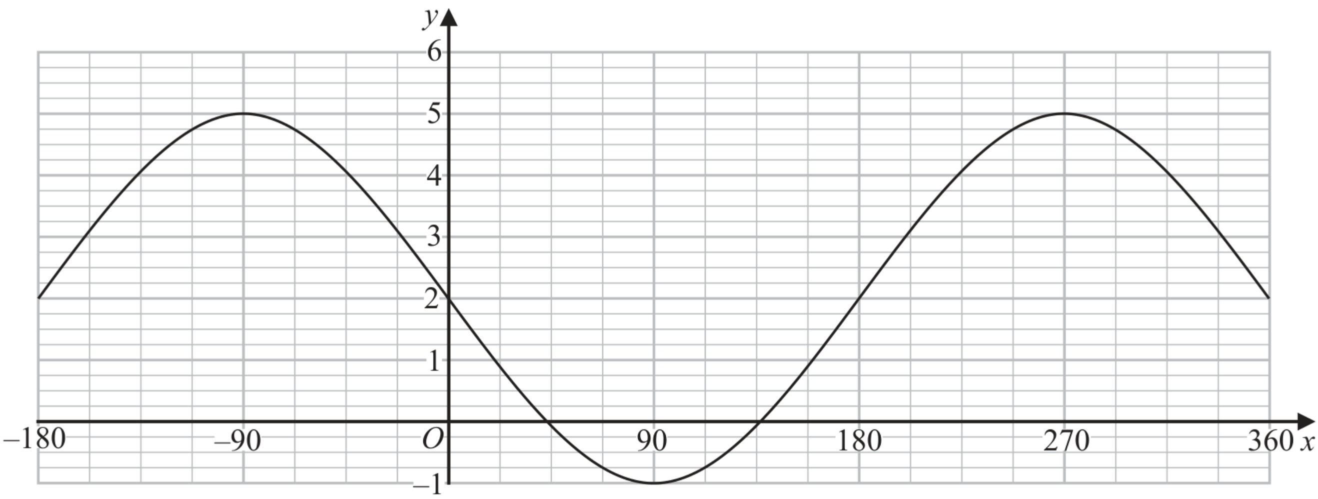
The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

....

(1)

The graph of $y = a\cos(x - b)° + c$ for $-180 \leqslant x \leqslant 360$ is drawn on the grid below.



(d) Find the value of a, the value of b and the value of c.

$a =$ ....

$b =$ ....

$c =$ ....

(3)

YOUR WORKING

(a) Find $\frac{dy}{dx}$

$\frac{dy}{dx} =$ ....

(2)

(b) Find the coordinates of the turning points on the graph with equation $y = 4x^3 + 5x^2 + 2x$
Show clear algebraic working.

....

(4)

(Total for Question 17 is 6 marks)

WORKING SPACE

4MA1-2H Q18 whole

4 marks

18 A particle is moving along a straight line that passes through the fixed point O
The displacement, s metres, of the particle from O at time t seconds is given by

$$s = 2t^3 - 5t^2 + 6t - 5$$

Find the value of t when the acceleration of the particle is 5 m/s²

$t =$ ....

WORKING SPACE

4MA1-1H Q20 M01

4 marks

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.
Show clear algebraic working.

....

WORKING SPACE

4MA1-1H Q24 **6 marks**

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**
The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3
Show clear algebraic working.

$y =$ ....

WORKING SPACE

4MA1-2H Q24 **5 marks**

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geqslant 0$, is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is a m/s^2

Find the value of a

$a =$ ....

WORKING SPACE

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x, in terms of p and m, for which the gradient of **C** is negative.

....

(Total for Question 23 is 4 marks)

YOUR WORKING

20 The radius of a right circular cylinder is x cm.

The height of the cylinder is $\left(\dfrac{800}{\pi x} - x\right)$ cm.

The volume of the cylinder is $V \text{ cm}^3$

Find the maximum value of V
Give your answer correct to the nearest whole number.

YOUR WORKING

January 2023 | 1H | Question 20 | 5 marks

Chapter OL-T38: Differentiation

(Total for Question 20 is 5 marks)

YOUR WORKING

4MA1-1H Q23 Q23

6 marks

23 A polygon has n sides, where $n > 5$

When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence.

Here are the first five terms of this sequence.

177 175 173 171 169

Find the value of n
Show clear algebraic working.

Question 23 continued

$n =$ ....

WORKING SPACE

4MA1-1H Q18 demand **5 marks**

18 The diagram shows the positions of three villages, A, B and C

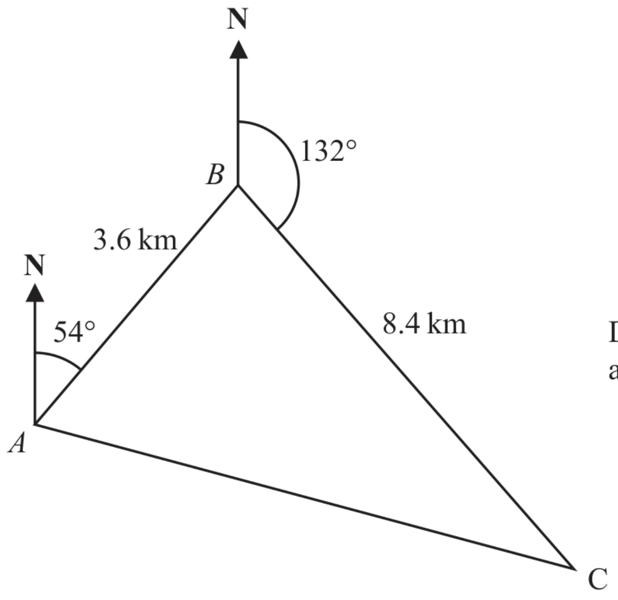


Diagram **NOT** accurately drawn

The bearing of B from A is 054°
The bearing of C from B is 132°

Melur walks from A to B
She then walks from B to C and from C to A

Melur walks at an average speed of 6 km/h

Work out the total time Melur takes.
Give your answer in hours and minutes.

.... hours .... minutes

WORKING SPACE

Continue your method **5 marks**

WORKING SPACE

4MA1-2H Q24 M01 **6 marks**

24 A, B and C are three points on horizontal ground.

$AB = 8.4$ metres $BC = 9.2$ metres

B is on a bearing of 067° from A
C is on a bearing of 129° from B

Calculate the bearing of A from C
Give your answer correct to the nearest degree.

4MA1-2H Q24 M01

6 marks

....

WORKING SPACE

16 The diagram shows the positions of three ships, A, B and C.

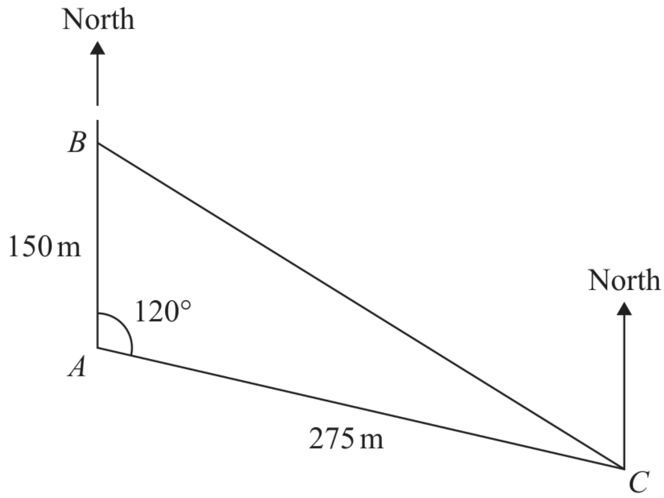


Diagram **NOT** accurately drawn

Ship B is due north of ship A.

The bearing of ship C from ship A is 120°

Calculate the bearing of ship C from ship B.
Give your answer correct to the nearest degree.

....°

YOUR WORKING

4MA1-1H Q22 demand **6 marks**

22 [In this question 1 cm = 1 unit on the x-axis and
1 cm = 1 unit on the y-axis]

P is a point on a circle with centre (0, 0)
The coordinates of P are (8, –10)

The line **L** is the tangent to the circle at the point P
L crosses the x-axis at the point Q and crosses the y-axis at the point R

Work out the length of RQ
Give your answer correct to 3 significant figures.

.... cm

WORKING SPACE

4MA1-2H Q16 M01, M02, M03, M04 **5 marks**

16 A, B, C and D are points on a circle, centre O

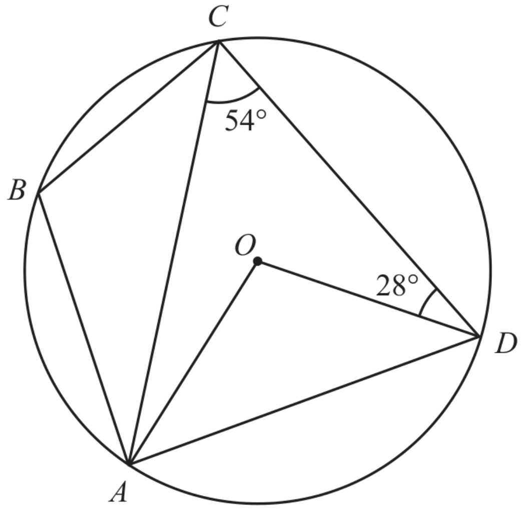


Diagram **NOT** accurately drawn

(a) (i) Work out the size of angle AOD

....

(1)

(ii) Give a reason for your answer to part (a)(i)

....

....

(1)

(b) Work out the size of angle CAO

....

(1)

(c) Work out the size of angle ABC

....

(2)

Continue your method **5 marks**

WORKING SPACE

4MA1-2H Q19 **3 marks**

19

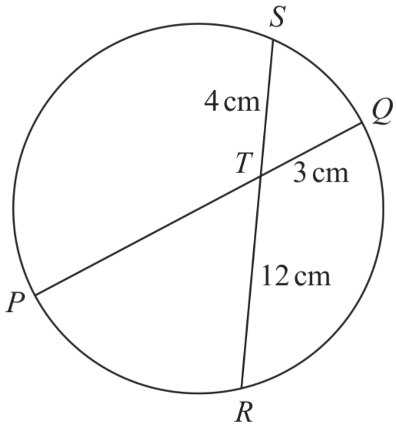


Diagram **NOT** accurately drawn

PTQ is a diameter of a circle.
RTS is a chord of the circle.

$TQ = 3$ cm $\quad ST = 4$ cm $\quad TR = 12$ cm

Calculate the radius of the circle.

.... cm

Continue your method **3 marks**

WORKING SPACE

4MA1-1H Q23 **5 marks**

23 A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

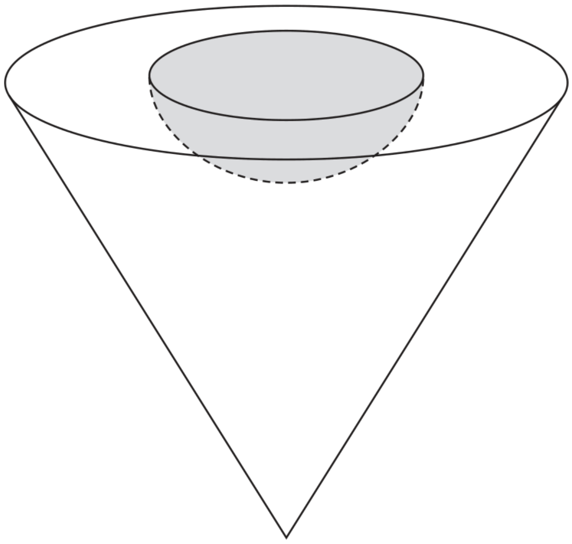


Diagram **NOT** accurately drawn

The radius of the hemisphere is $2x$ cm
The radius of the base of the cone is $5x$ cm
The vertical height of the cone is $6x$ cm

The volume of the solid shape is 6948π cm^3

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.
Give your answer correct to the nearest integer.

.... cm^2

Continue your method **5 marks**

WORKING SPACE

4MA1-1H Q21 demand **5 marks**

21 Here is a triangular prism *ABCDEF*

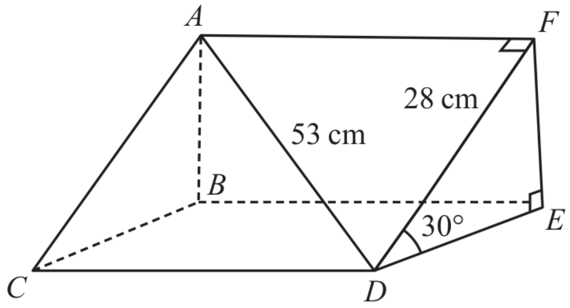


Diagram **NOT** accurately drawn

$AD = 53$ cm
$DF = 28$ cm
Angle $FDE = 30°$

Work out the volume of the triangular prism.
Give your answer correct to the nearest whole number.

.... cm^3

WORKING SPACE

4MA1-1H Q22 Q22 **5 marks**

22 A solid is made from a cone and a hemisphere.

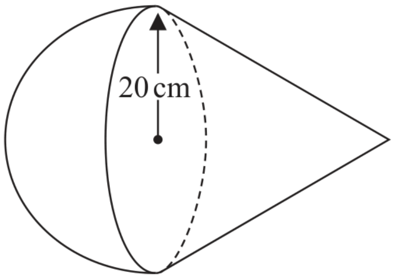


Diagram **NOT** accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm.

The curved surface area of the cone is $580\pi \text{ cm}^2$

The volume of the solid is $k\pi \text{ cm}^3$

Work out the exact value of k

$k =$ ....

Continue your method **5 marks**

WORKING SPACE

26 Here is a sector, *AOB*, of a circle with centre *O* and angle *AOB* = $x°$

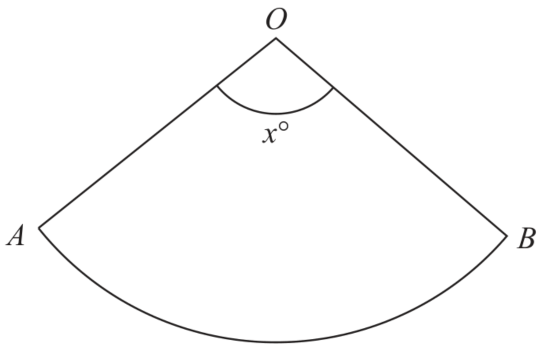


Diagram **NOT** accurately drawn

The sector can form the curved surface of a cone by joining *OA* to *OB*.

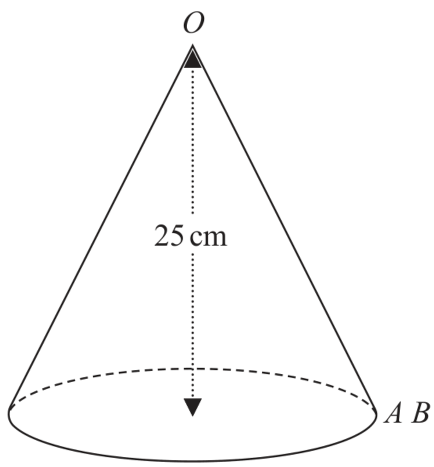


Diagram **NOT** accurately drawn

The height of the cone is 25 cm.
The volume of the cone is 1600 cm^3

Work out the value of *x*.
Give your answer correct to the nearest whole number.

$x =$ ....

YOUR WORKING

17 The diagram shows a solid prism *ABCDEFGH*.

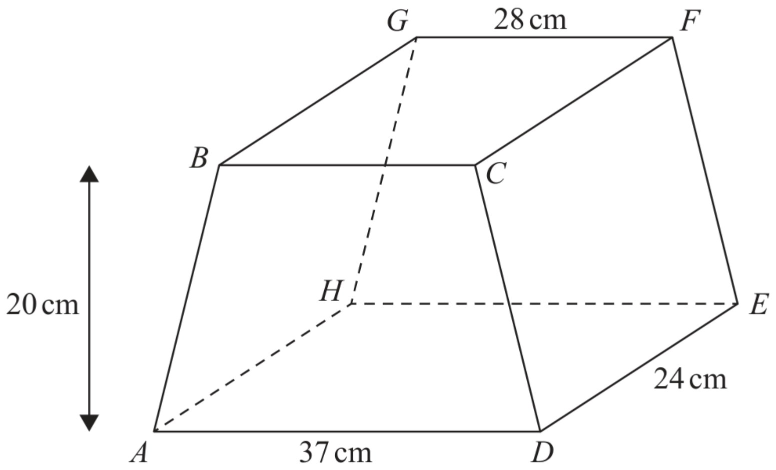


Diagram **NOT** accurately drawn

The trapezium *ABCD*, in which *AD* is parallel to *BC*, is a cross section of the prism.
The base *ADEH* of the prism is a horizontal plane.
ADEH and *BCFG* are rectangles.
The midpoint of *BC* is vertically above the midpoint of *AD* so that *BA* = *CD*.

AD = 37 cm *GF* = 28 cm *DE* = 24 cm

The perpendicular distance between edges *AD* and *BC* is 20 cm.

(a) Work out the total surface area of the prism.

.... cm^2

(4)

YOUR WORKING

January 2023 | 2H | Question 21 | 3 marks

Chapter OL-T45: Volume & Surface Area

21 Given that the surface area of a sphere is $49\pi\,\text{cm}^2$

find the volume of the sphere.
Give your answer correct to the nearest integer.

.... cm^3

(Total for Question 21 is 3 marks)

YOUR WORKING

The height of solid **A** is 31 cm
The height of solid **B** is 18.6 cm

Given that

volume of solid **A** – volume of solid **B** $= 735\,\text{cm}^3$

find the volume of solid **A**

.... cm^3

(Total for Question 26 is 4 marks)

WORKING SPACE

4MA1-1H Q23 demand **5 marks**

23 Solid **A** is similar to solid **B**

Here is some information about solid **A** and solid **B**

	solid **A**	solid **B**
Height (cm)	3^x	
Area (cm^2)	7776	486
Volume (cm^3)	8^x	2^{x+4}

Work out the height of solid **B**
Give your answer as a decimal.

.... cm

WORKING SPACE

4MA1-2H Q23 whole **5 marks**

23 Here is a frustum of a cone.
The frustum is made by removing a small cone from a similar large cone.

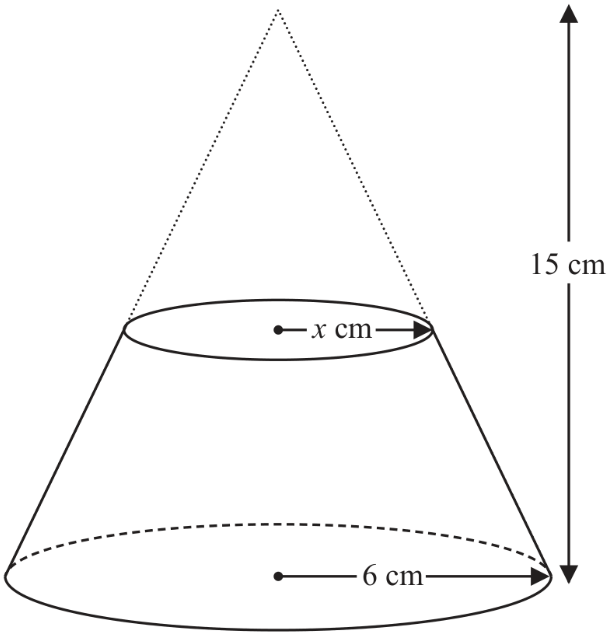


Diagram **NOT** accurately drawn

The height of the large cone is 15 cm.

The radius of the base of the large cone is 6 cm.
The radius of the base of the small cone is x cm.

Given that the volume of the frustum is $\frac{4212}{25}\pi$ cm^3

work out the value of x
Show clear algebraic working.

$x =$ ....

WORKING SPACE

Continue your method **5 marks**

WORKING SPACE

4MA1-1H Q17 **4 marks**

17 **A** and **B** are two similar vases.

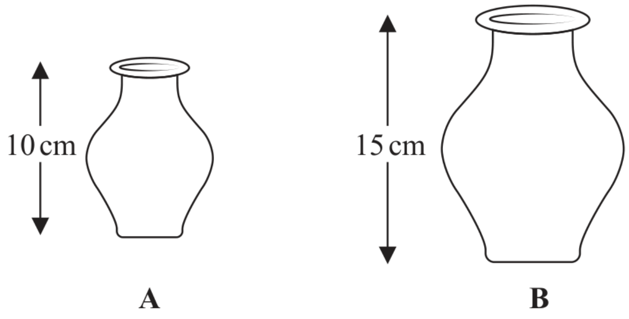


Diagram **NOT** accurately drawn

Vase **A** has height 10 cm.
Vase **B** has height 15 cm.

The difference between the volume of vase **A** and the volume of vase **B** is 1197 cm^3

Calculate the volume of vase **A**

.... cm^3

Continue your method **4 marks**

WORKING SPACE

January 2023 | 2H | Question 19 | 4 marks

Chapter OL-T47: Area & Volume of Similar Shapes

19 **A** and **B** are two similar vases.

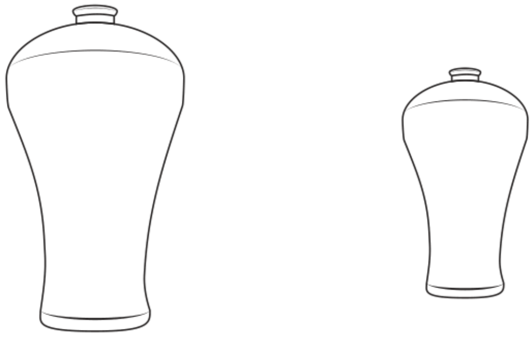


Diagram **NOT** accurately drawn

A **B**

The vases are such that

$$\text{surface area of vase } \mathbf{B} = \frac{25}{64} \times \text{surface area of vase } \mathbf{A}$$

and that

$$\text{volume of vase } \mathbf{A} - \text{volume of vase } \mathbf{B} = 541.8\,\text{cm}^3$$

Calculate the volume of vase **B**

.... cm^3

(Total for Question 19 is 4 marks)

YOUR WORKING

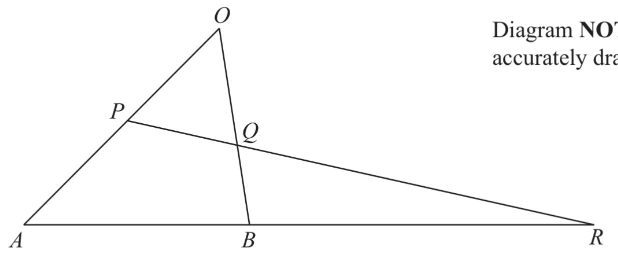


Diagram **NOT** accurately drawn

OAB is a triangle.
P is the midpoint of OA
Q is a point on OB

ABR and PQR are straight lines.

$\overrightarrow{OA} = 12\mathbf{a}$ $\overrightarrow{OB} = 8\mathbf{b}$

(a) Express $\overrightarrow{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

....
(1)

WORKING SPACE

4MA1-1H Q23

23

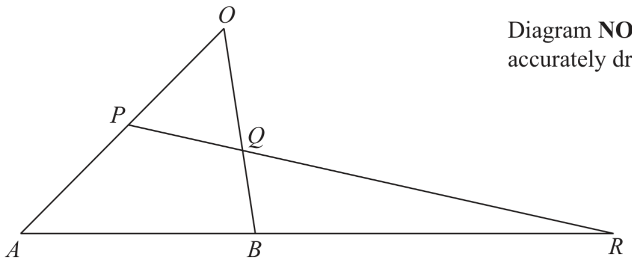


Diagram **NOT** accurately drawn

OAB is a triangle.
P is the midpoint of *OA*
Q is a point on *OB*

ABR and *PQR* are straight lines.

$\overrightarrow{OA} = 12\mathbf{a}$ $\overrightarrow{OB} = 8\mathbf{b}$

$AB : BR = 1 : 2$ $\overrightarrow{OQ} = n\mathbf{b}$

(b) Use a vector method to find the value of n

$n =$ ....

(4)

WORKING SPACE

4MA1-2H Q24 **6 marks**

24

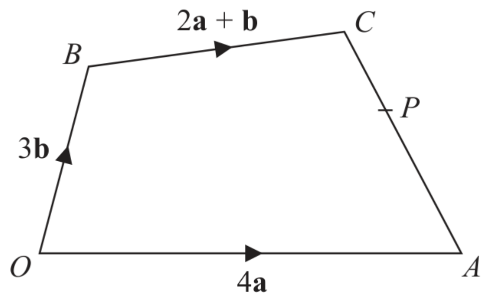


Diagram **NOT** accurately drawn

The diagram shows a quadrilateral *OACB* in which

$\overrightarrow{OA} = 4\mathbf{a}$ $\overrightarrow{OB} = 3\mathbf{b}$ $\overrightarrow{BC} = 2\mathbf{a} + \mathbf{b}$

(a) Find $\overrightarrow{AC}$ in terms of **a** and **b**
Give your answer in its simplest form.

$\overrightarrow{AC}$ = ...

(2)

WORKING SPACE

4MA1-2H Q24 **6 marks**

24

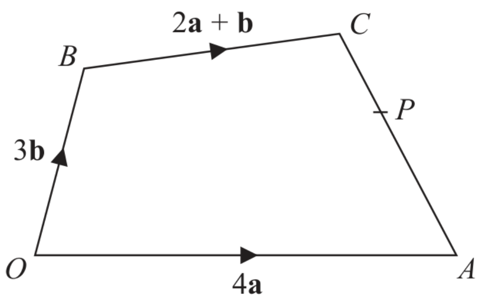


Diagram **NOT** accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$\overrightarrow{OA} = 4\mathbf{a}$ $\quad\overrightarrow{OB} = 3\mathbf{b}$ $\quad\overrightarrow{BC} = 2\mathbf{a} + \mathbf{b}$

The point P lies on AC such that $AP : PC = 3 : 2$

The point Q is such that OPQ and BCQ are straight lines.

(b) Using a vector method, find $\overrightarrow{OQ}$ in terms of **a** and **b**
Give your answer in its simplest form.
Show your working clearly.

$\overrightarrow{OQ} =$ ....
(4)

Continue your method **6 marks**

WORKING SPACE

4MA1-2H Q22 (a)(i), (a)(ii), (b) **5 marks**

22

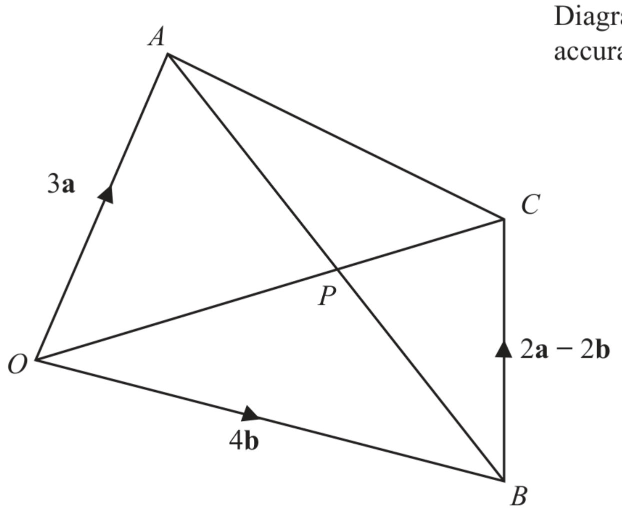


Diagram **NOT** accurately drawn

$OACB$ is a quadrilateral.

$\overrightarrow{OA} = 3\mathbf{a}$ $\overrightarrow{OB} = 4\mathbf{b}$ $\overrightarrow{BC} = 2\mathbf{a} - 2\mathbf{b}$

(a) (i) Find the vector $\overrightarrow{OC}$ in terms of **a** and **b**
Simplify your answer.

$\overrightarrow{OC} =$ ....
(1)

(ii) Find the vector $\overrightarrow{AB}$ in terms of **a** and **b**

$\overrightarrow{AB} =$ ....
(1)

The point P lies on AB and on OC

(b) Using a vector method, find the ratio $AP : PB$
Show your working clearly.

....
(3)

Continue your method **5 marks**

WORKING SPACE

4MA1-1H Q19 **5 marks**

19

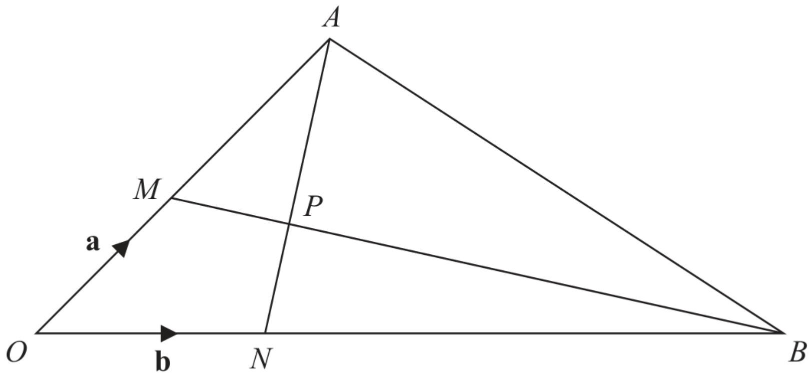


Diagram **NOT** accurately drawn

OMA, *ONB*, *MPB* and *NPA* are straight lines.

M is the midpoint of *OA*

$ON : NB = 1 : 5$

$\overrightarrow{OM} = \mathbf{a}$ $\overrightarrow{ON} = \mathbf{b}$

(a) Find in terms of **a** and **b** the vector $\overrightarrow{AN}$

....

(1)

(b) Use a vector method to find the ratio $AP : PN$

$AP : PN =$ ....

(4)

Continue your method **5 marks**

WORKING SPACE

4MA1-1H Q24 M01, M02, M03 **6 marks**

24 OAB is a triangle.

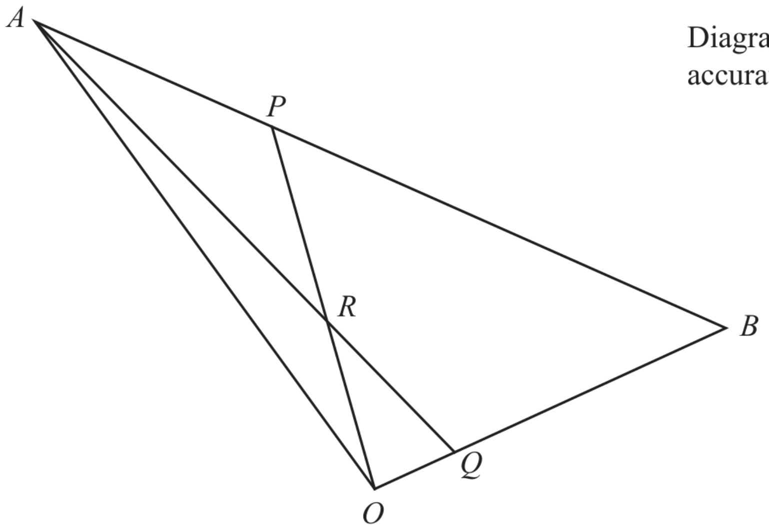


Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 10\mathbf{a}$ $\overrightarrow{OB} = 10\mathbf{b}$

ARQ and ORP are straight lines.

$\overrightarrow{AP} = \frac{1}{4}\overrightarrow{AB}$ and $\overrightarrow{OQ} = \frac{1}{5}\overrightarrow{OB}$

Write the following vectors in terms of **a** and **b**
Simplify your answers.

(i) $\overrightarrow{AQ}$

....
(1)

(ii) $\overrightarrow{OP}$

....
(1)

4MA1-1H Q24 M01, M02, M03

6 marks

(iii) $\overrightarrow{OR}$

....

(4)

WORKING SPACE

4MA1-2H Q17 M01 **3 marks**

17 Here are two vectors.

$$\overrightarrow{FG} = \begin{pmatrix} -5 \\ 2 \end{pmatrix} \qquad \overrightarrow{HG} = \begin{pmatrix} 4 \\ 14 \end{pmatrix}$$

Calculate the magnitude of the vector $\overrightarrow{HF}$

....

WORKING SPACE

19 OAB is a triangle.

$\overrightarrow{OA} = \mathbf{a}$ $\overrightarrow{OB} = \mathbf{b}$

The point C lies on OA such that $OC : CA = 1 : 2$
The point D lies on OB such that $OD : DB = 1 : 2$

Using a vector method, prove that $ABDC$ is a trapezium.

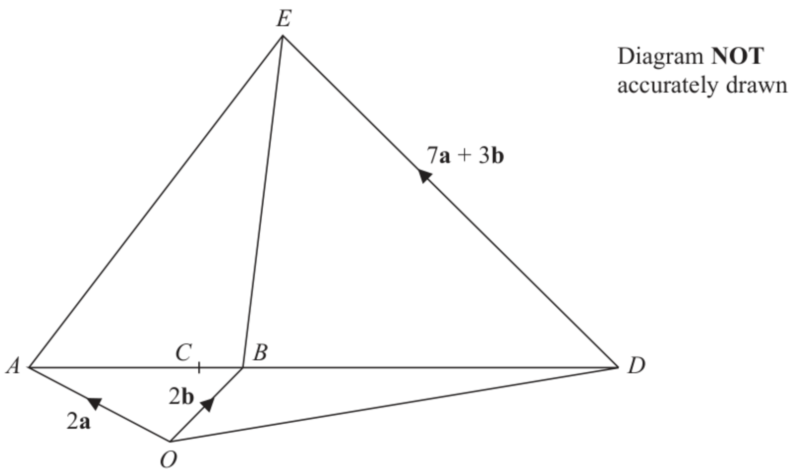
(Total for Question 19 is 3 marks)

YOUR WORKING

January 2023 | 1H | Question 24 | 5 marks

Chapter OL-T51: Vectors

24 *OAED* is a quadrilateral.



$\overrightarrow{OA} = 2\mathbf{a}$ $\overrightarrow{OB} = 2\mathbf{b}$ $\overrightarrow{DE} = 7\mathbf{a} + 3\mathbf{b}$

$AB : BD = 1 : 2$

The point C on AB is such that OCE is a straight line.

Use a vector method to find the ratio of $OC : CE$

YOUR WORKING

January 2023 | 1H | Question 24 | 5 marks

Chapter OL-T51: Vectors

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(Total for Question 24 is 5 marks)

YOUR WORKING